Grade 11 – Term 1
C8: Student's t in Context

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1 Models and calculation tools

Open the normal and Student's t reference tables. Page 1 gives $\Phi(z) = P(Z \leq z)$ for positive z on a hundredth grid (through 2.59). Page 2 gives positive t critical values for upper-tail probabilities $0.30, 0.20, 0.10, 0.05, 0.025, 0.01, 0.005$, for df 1–30 and selected larger df. Those entries do not give arbitrary CDF values. Use a suitable calculator's t CDF and inverse t CDF for this activity; normal CDF and inverse normal are also allowed. Numerical CDF and quantile results in the solutions are supplementary to the tables. Never turn table bounds into a single probability or error.

Write $F_\nu(t) = P(T \leq t)$ for $T \sim t_\nu$. Symmetry gives $F_\nu(-c) = 1 - F_\nu(c)$; a right tail is $1 - F_\nu(c)$. An interval has probability $F_\nu(b) - F_\nu(a)$ and its outside complement has probability $F_\nu(a) + 1 - F_\nu(b)$. Replace F_ν by Φ for normal calculations. Endpoints have zero probability. For central area p, each tail has $(1-p)/2$; for combined symmetric tails q, each tail has $q/2$.

A contextual location-scale model is explicitly $X = m + aT$, with $a > 0$ a scale parameter; a is not automatically the standard deviation. Thus $T = (X - m)/a$. These are stipulated models for signed deviations, not claims that arbitrary individual measurements follow t. For independent random observations from a normal population, with unknown population standard deviation, the sampling statistic instead is

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Here $\bar{X}$ and S vary across samples; do not treat observed s as a fixed population standard deviation.

Attempt each task before checking the worked solution. Calculate with full precision, then report probabilities to four decimals, standardized quantiles to three decimals, contextual values to two decimals, and percentage errors to two decimals. Displayed rounded intermediate values are not the inputs used for errors. Diagrams shade the requested region; curves continue beyond the displayed range. **C8.** Calculate contextual t probabilities and find cutoffs from target probabilities.

2 Direct probability problems

Problems:

2.1. Delivery schedule deviations
2.2. Production cost deviations
2.3. Electricity forecast deviations
2.4. Monthly sales deviations
2.5. Commute forecast deviations
2.6. Revenue planning alerts

2.1 Delivery schedule deviations

Problem. A planning team models a delivery time minus its scheduled time, measured in minutes. Use $T \sim t_9$ and $X = 0 + 5T$. The scale is 5 minutes.

Questions/Tasks

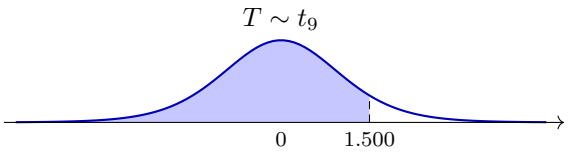
C8 Find $P(X < 7.5)$, and $P(X < -6)$. Show the standardized event, explain symmetry or complements as needed, and interpret each answer.

Worked solution

Standardize each boundary using $(x - 0)/5$.

Part 1.

$$P(X < 7.5) = P(T < 1.5).$$



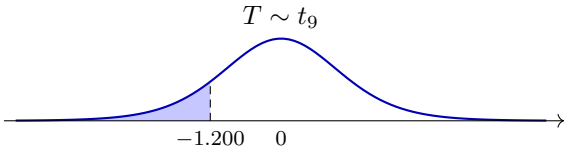
For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. The region is cumulative area to the left. Supplementary CDF values: $F_9(1.5) \approx 0.9160747$.

$$P(T < 1.5) = F_9(1.5) \approx 0.9161.$$

Interpretation. About 91.61% of delivery schedule deviations satisfy the stated condition. Negative deviations mean outcomes below the schedule, budget, forecast or target.

Part 2.

$$P(X < -6) = P(T < -1.2).$$



For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. The region is cumulative area to the left. By symmetry, $F_9(-1.2) = 1 - F_9(1.2) \approx 1 - 0.8696134 = 0.1303866$. Supplementary CDF values: $F_9(-1.2) \approx 0.1303866$.

$$P(T < -1.2) = F_9(-1.2) \approx 0.1304.$$

Interpretation. About 13.04% of delivery schedule deviations satisfy the stated condition. Negative deviations mean outcomes below the schedule, budget, forecast or target.

2.2 Production cost deviations

Problem. A planning team models a batch cost minus its budget, measured in monetary units. Use $T \sim t_{14}$ and $X = 0 + 20T$. The scale is 20 monetary units.

Questions/Tasks

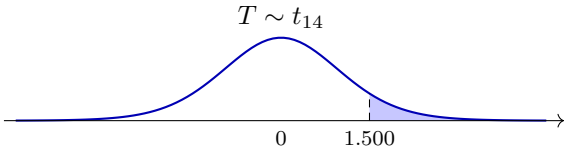
C8 Find $P(X > 30)$, and $P(X > -16)$. Show the standardized event, explain symmetry or complements as needed, and interpret each answer.

Worked solution

Standardize each boundary using $(x - 0)/20$.

Part 1.

$$P(X > 30) = P(T > 1.5).$$



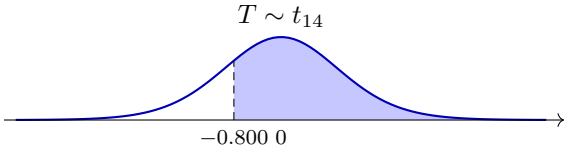
For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Take the complement of the cumulative area. Supplementary CDF values: $F_{14}(1.5) \approx 0.9220873$.

$$P(T > 1.5) = 1 - F_{14}(1.5) \approx 0.0779.$$

Interpretation. About 7.79% of production cost deviations satisfy the stated condition. Negative deviations mean outcomes below the schedule, budget, forecast or target.

Part 2.

$$P(X > -16) = P(T > -0.8).$$



For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Take the complement of the cumulative area. By symmetry, $F_{14}(-0.8) = 1 - F_{14}(0.8) \approx 1 - 0.7814622 = 0.2185378$. Supplementary CDF values: $F_{14}(-0.8) \approx 0.2185378$.

$$P(T > -0.8) = 1 - F_{14}(-0.8) \approx 0.7815.$$

Interpretation. About 78.15% of production cost deviations satisfy the stated condition. Negative deviations mean outcomes below the schedule, budget, forecast or target.

2.3 Electricity forecast deviations

Problem. A planning team models household use minus its forecast, measured in kWh. Use $T \sim t_{19}$ and $X = 0 + 50T$. The scale is 50 kWh.

Questions/Tasks

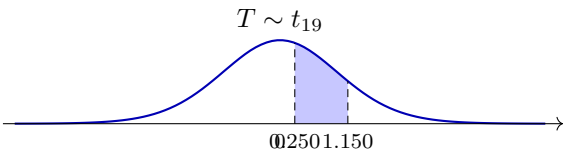
C8 Find $P(12.5 < X < 57.5)$. Show the standardized event, explain symmetry or complements as needed, and interpret each answer.

Worked solution

Standardize each boundary using $(x - 0)/50$.

Part 1.

$$P(12.5 < X < 57.5) = P(0.25 < T < 1.15).$$



For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Subtract the left cumulative area from the right cumulative area. Supplementary CDF values: $F_{19}(0.25) \approx 0.5973648$, $F_{19}(1.15) \approx 0.8677933$.

$$P(0.25 < T < 1.15) = F_{19}(1.15) - F_{19}(0.25) \approx 0.2704.$$

Interpretation. About 27.04% of electricity forecast deviations satisfy the stated condition. Negative deviations mean outcomes below the schedule, budget, forecast or target.

2.4 Monthly sales deviations

Problem. A planning team models monthly sales minus their target, measured in units. Use $T \sim t_{11}$ and $X = 0 + 20T$. The scale is 20 units.

Questions/Tasks

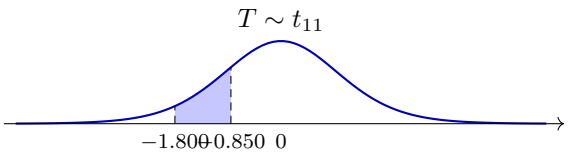
C8 Find $P(-36 < X < -17)$. Show the standardized event, explain symmetry or complements as needed, and interpret each answer.

Worked solution

Standardize each boundary using $(x - 0)/20$.

Part 1.

$$P(-36 < X < -17) = P(-1.8 < T < -0.85).$$



For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Subtract the left cumulative area from the right cumulative area. By symmetry, $F_{11}(-1.8) = 1 - F_{11}(1.8) \approx 1 - 0.9503421 = 0.0496579$. By symmetry, $F_{11}(-0.85) = 1 - F_{11}(0.85) \approx 1 - 0.7932783 = 0.2067217$. Supplementary CDF values: $F_{11}(-1.8) \approx 0.0496579$, $F_{11}(-0.85) \approx 0.2067217$.

$$P(-1.8 < T < -0.85) = F_{11}(-0.85) - F_{11}(-1.8) \approx 0.1571.$$

Interpretation. About 15.71% of monthly sales deviations satisfy the stated condition. Negative deviations mean outcomes below the schedule, budget, forecast or target.

2.5 Commute forecast deviations

Problem. A planning team models a commute time minus its forecast, measured in minutes. Use $T \sim t_{24}$ and $X = 0 + 10T$. The scale is 10 minutes.

Questions/Tasks

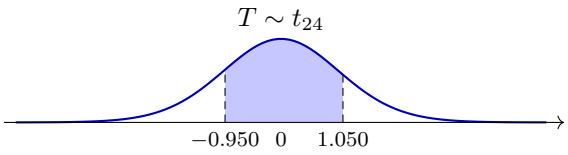
C8 Find $P(-9.5 < X < 10.5)$. Show the standardized event, explain symmetry or complements as needed, and interpret each answer.

Worked solution

Standardize each boundary using $(x - 0)/10$.

Part 1.

$$P(-9.5 < X < 10.5) = P(-0.95 < T < 1.05).$$



For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Subtract the left cumulative area from the right cumulative area. By symmetry, $F_{24}(-0.95) = 1 - F_{24}(0.95) \approx 1 - 0.8242092 = 0.1757908$. Supplementary CDF values: $F_{24}(-0.95) \approx 0.1757908$, $F_{24}(1.05) \approx 0.8479116$.

$$P(-0.95 < T < 1.05) = F_{24}(1.05) - F_{24}(-0.95) \approx 0.6721.$$

Interpretation. About 67.21% of commute forecast deviations satisfy the stated condition. Negative deviations mean outcomes below the schedule, budget, forecast or target.

2.6 Revenue planning alerts

Problem. A planning team models weekly revenue minus its plan, measured in thousand monetary units. Use $T \sim t_5$ and $X = 0 + 8T$. The scale is 8 thousand monetary units.

Questions/Tasks

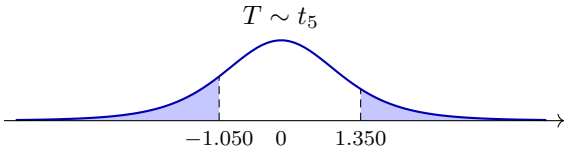
C8 Find $P(X < -8.4 \text{ or } X > 10.8)$, and $P(X < -16 \text{ or } X > 16)$. Show the standardized event, explain symmetry or complements as needed, and interpret each answer.

Worked solution

Standardize each boundary using $(x - 0)/8$.

Part 1.

$$P(X < -8.4 \text{ or } X > 10.8) = P(T < -1.05 \text{ or } T > 1.35).$$



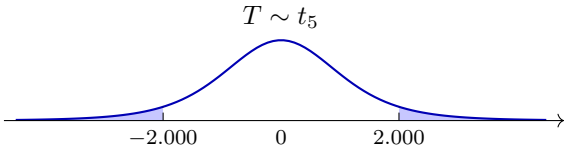
For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Add the disjoint outside regions, or subtract the intervening interval from one. By symmetry, $F_5(-1.05) = 1 - F_5(1.05) \approx 1 - 0.8291023 = 0.1708977$. Supplementary CDF values: $F_5(-1.05) \approx 0.1708977$, $F_5(1.35) \approx 0.8825456$.

$$P(T < -1.05 \text{ or } T > 1.35) = F_5(-1.05) + 1 - F_5(1.35) \approx 0.2884.$$

Interpretation. About 28.84% of revenue planning alerts satisfy the stated condition. Negative deviations mean outcomes below the schedule, budget, forecast or target.

Part 2.

$$P(X < -16 \text{ or } X > 16) = P(T < -2 \text{ or } T > 2).$$



For a negative boundary, use $F_\nu(-c) = 1 - F_\nu(c)$. Add the disjoint outside regions, or subtract the intervening interval from one. By symmetry, $F_5(-2) = 1 - F_5(2) \approx 1 - 0.9490303 = 0.0509697$. Supplementary CDF values: $F_5(-2) \approx 0.0509697$, $F_5(2) \approx 0.9490303$.

$$P(T < -2 \text{ or } T > 2) = F_5(-2) + 1 - F_5(2) \approx 0.1019.$$

Interpretation. About 10.19% of revenue planning alerts satisfy the stated condition. Negative deviations mean outcomes below the schedule, budget, forecast or target.

3 Inverse probability problems

Problems:

3.1. Top delivery delays

3.2. Bottom cost deviations

3.3. Central energy deviations

3.4. Central sales deviations

3.5. Two-tail checks on filling weights

3.6. Two-tail checks on processing times

3.1 Top delivery delays

Problem. Model delivery time minus its schedule in minutes by $X_T = 0 + 5T$, $T \sim t_9$. Find the cutoff exceeded by the top 5%.

Questions/Tasks

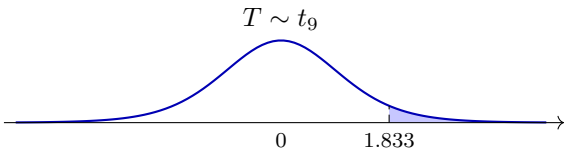
C8 Write the target event, convert to cumulative probabilities, find the standardized cutoff(s) and convert to contextual units. Explain the tail conversion and interpret the result.

Worked solution

$$P(X_T > x) = 0.0500 \quad \Longrightarrow \quad P(T \le c) = 0.9500, \qquad c = \frac{x - 0}{5}.$$

A top percentage is a right tail, so subtract it from one before inverse CDF.

$$c_T = F_9^{-1}(0.95) \approx 1.833.$$



$$x_T = 0 + 5c_T \approx 9.17 \text{ minutes.}$$

Interpretation. About 5% of deviations are above 9.17 minutes.

3.2 Bottom cost deviations

Problem. Model batch cost minus its budget in monetary units by $X_T = 0 + 20T$, $T \sim t_{11}$. Find the cutoff below which the bottom 10% fall.

Questions/Tasks

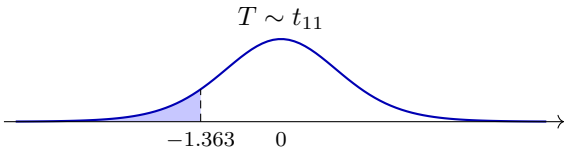
C8 Write the target event, convert to cumulative probabilities, find the standardized cutoff(s) and convert to contextual units. Explain the tail conversion and interpret the result.

Worked solution

$$P(X_T < x) = 0.1000 \quad \Longrightarrow \quad P(T \le c) = 0.1000, \qquad c = \frac{x - 0}{20}.$$

The bottom percentage is already a left cumulative probability; the cutoff is negative by symmetry.

$$c_T = F_{11}^{-1}(0.1) \approx -1.363.$$



$$x_T = 0 + 20c_T \approx -27.27 \text{ monetary units.}$$

Interpretation. About 10% of deviations are below -27.27 monetary units.

3.3 Central energy deviations

Problem. Model electricity use minus its forecast in kWh by $X_T = 0 + 50T$, $T \sim t_{19}$. Find the symmetric interval about 0 containing the middle 95%.

Questions/Tasks

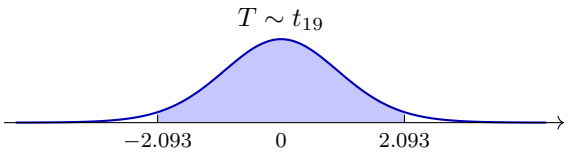
C8 Write the target event, convert to cumulative probabilities, find the standardized cutoff(s) and convert to contextual units. Explain the tail conversion and interpret the result.

Worked solution

$$P(-c < T < c) = 0.9500 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.0250.$$

Divide the outside complement $1 - p$ by two because symmetry makes the tails equal.

$$F_{19}(c) = 0.9750, \qquad c_T = F_{19}^{-1}(0.975) \approx 2.093.$$



The standardized endpoints are $[-2.093, 2.093]$.

$$L_T = 0 - 50c_T \approx -104.65, \qquad U_T = 0 + 50c_T \approx 104.65 \quad \text{(kWh)}.$$

Interpretation. The t interval contains 95% of modeled deviations; 2.5% lie in each excluded tail.

3.4 Central sales deviations

Problem. Model sales minus their target in units by $X_T = 0 + 20T$, $T \sim t_{29}$. Find the symmetric interval about 0 containing the middle 80%.

Questions/Tasks

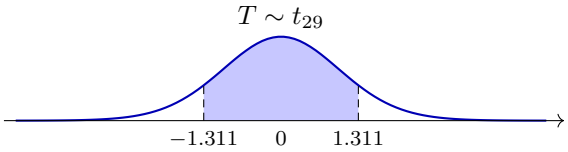
C8 Write the target event, convert to cumulative probabilities, find the standardized cutoff(s) and convert to contextual units. Explain the tail conversion and interpret the result.

Worked solution

$$P(-c < T < c) = 0.8000 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.1000.$$

Divide the outside complement $1 - p$ by two because symmetry makes the tails equal.

$$F_{29}(c) = 0.9000, \qquad c_T = F_{29}^{-1}(0.9) \approx 1.311.$$



The standardized endpoints are $[-1.311, 1.311]$.

$$L_T = 0 - 20c_T \approx -26.23, \qquad U_T = 0 + 20c_T \approx 26.23 \quad \text{(units)}.$$

Interpretation. The t interval contains 80% of modeled deviations; 10% lie in each excluded tail.

3.5 Two-tail checks on filling weights

Problem. A quality team takes independent random samples of $n = 6$ filling weights from a normal population with mean μ and unknown standard deviation. Define $T = (\bar{X} - \mu)/(S/\sqrt{6}) \sim t_5$. Find symmetric limits for the standardized sample mean leaving a combined two-tail probability of 0.1000.

Questions/Tasks

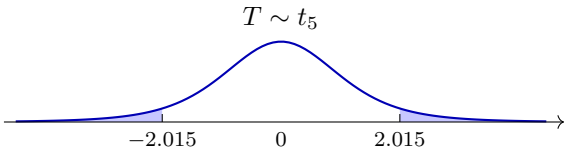
C8 Write the target event, convert to cumulative probabilities, find the standardized cutoff(s). Explain the tail conversion and interpret the result.

Worked solution

$$P(|T| > c) = 0.1000 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.0500.$$

Divide the combined outside area by two because symmetry makes the tails equal.

$$F_5(c) = 0.9500, \qquad c_T = F_5^{-1}(0.95) \approx 2.015.$$



The standardized endpoints are $[-2.015, 2.015]$. **Interpretation.** The t interval contains 90% of standardized sample-mean outcomes; 5% lie in each excluded tail.

3.6 Two-tail checks on processing times

Problem. A quality team takes independent random samples of $n = 25$ processing times from a normal population with mean μ and unknown standard deviation. Define $T = (\bar{X} - \mu)/(S/\sqrt{25}) \sim t_{24}$. Find symmetric limits for the standardized sample mean leaving a combined two-tail probability of 0.0400.

Questions/Tasks

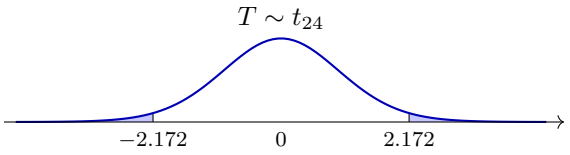
C8 Write the target event, convert to cumulative probabilities, find the standardized cutoff(s). Explain the tail conversion and interpret the result.

Worked solution

$$P(|T| > c) = 0.0400 \quad \Longrightarrow \quad P(T > c) = P(T < -c) = 0.0200.$$

Divide the combined outside area by two because symmetry makes the tails equal.

$$F_{24}(c) = 0.9800, \qquad c_T = F_{24}^{-1}(0.98) \approx 2.172.$$



The standardized endpoints are $[-2.172, 2.172]$. **Interpretation.** The t interval contains 96% of standardized sample-mean outcomes; 2% lie in each excluded tail.